

# The Universal Thin Filament Code (uTFC): Unifying Cellular Automata and Sturm-Liouville Solvers via the Wolf Parameterization

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## Abstract

We propose a unified computational framework that integrates the discrete algebraic evolution of Cellular Automata (CA) with the continuous stability analysis of the Thin Filament Code (TFC). By identifying the *Stiff Rule 30* update mechanism with the Wolf Parameterization of magnetospheric flux tubes, we derive a **Universal TFC Solver**. We prove that the statistical moments of the CA lattice converge to the continuous background profiles required by the Sturm-Liouville stability equation. We demonstrate that the transition from Sedenion Chaos ( $\beta \rightarrow 0$ ) to Octonionic Order ( $\beta \approx 1.91$ ) is isomorphic to the stabilization of the TFC eigenmodes. We present numerical results showing that the *Standard Model* of particle physics emerges as the critical attractor of this computational phase space.

## 1 Introduction: The Discrete-Continuum Bridge

Physics is traditionally divided into discrete models (Lattice Gauge Theory, Cellular Automata) and continuous models (Differential Geometry, MHD). The Axiomatic Physical Homeostasis (APH) framework suggests these are dual descriptions of the same underlying algebraic reality.

- **The Micro-State:** *Rule 30* represents the propagation of non-associative defects (Sedenion Zero Divisors) in a discrete lattice.
- **The Macro-State:** The Thin Filament Code (TFC) solves for the stability of flux tubes in a continuous magnetized plasma [3].

We propose that the *Wolf Parameterization*—the specific profiles of Pressure  $P(s)$ , Magnetic Field  $B(s)$ , and Density  $\rho(s)$  used in TFC—is the continuum limit of the weighted average of CA cells surviving the *Stiffness Filter*.

## 2 Mathematical Formulation

### 2.1 The Stiff Rule 30 Automaton

Let  $\mathcal{L}^{(t)} = \{s_i^{(t)}\}$  be a 1D lattice where  $s_i \in \{0, 1\}$ . Standard Rule 30 is given by the map  $\Phi_{30} : \{0, 1\}^3 \rightarrow \{0, 1\}$ . We introduce the **Stiffness Filter**  $\Theta_\beta$ . In the APH framework, a cell represents a local algebraic product. If the product is non-associative, it generates an *Associator Error*  $\mathcal{A}$ .

$$s_i^{(t+1)} = \Phi_{30}(s_{i-1}, s_i, s_{i+1}) \cdot \Theta(\beta - \mathcal{A}_{local}) \quad (1)$$

where  $\mathcal{A}_{local}$  is a stochastic variable drawn from the zero-divisor distribution of the algebra  $\mathbb{A}_n$ . For simulation purposes, this is equivalent to a probabilistic kill switch:

$$P(\text{survival}) = 1 - e^{-\beta} \quad (2)$$

## 2.2 The Wolf Continuum Mapping

We define the mapping  $\mathcal{M} : \mathcal{L} \rightarrow C^\infty(\mathbb{R})$  that transforms the discrete lattice into continuous TFC profiles. Let  $s$  be the coordinate along the flux tube (or time in the CA). We define the *Wolf Density*  $\rho(s)$  and *Wolf Pressure*  $P(s)$  as rolling statistical moments of the lattice generation  $t \leftrightarrow s$ .

**Definition 1** (Wolf Density). *The continuous inertial density is the ensemble average of active bits:*

$$\rho(s) = \lim_{N \rightarrow \infty} \frac{1}{N} \sum_{i=1}^N s_i^{(t=s)} \quad (3)$$

**Definition 2** (Wolf Pressure). *The restoring pressure is inversely proportional to the local variance (entropy) of the lattice, scaled by the stiffness  $\beta$ :*

$$P(s) = \beta \cdot \left( \text{Var}[s_i^{(t=s)}] + \epsilon \right)^{-1} \quad (4)$$

## 2.3 The Universal Stability Theorem

The TFC determines stability by solving the Sturm-Liouville equation for the displacement  $\xi$ :

$$\frac{d}{ds} \left( P(s) \frac{d\xi}{ds} \right) + \rho(s) \omega^2 \xi = 0 \quad (5)$$

**Theorem 1** (The Isomorphism of Stability). *The Lyapunov exponent  $\lambda$  of the Cellular Automaton is isomorphic to the imaginary part of the TFC eigenfrequency  $\omega$ .*

- **Chaos** ( $\lambda > 0$ ): Corresponds to  $\omega^2 < 0$  (Imaginary  $\omega$ ). The TFC flux tube is unstable (Interchange Instability).
- **Order** ( $\lambda \leq 0$ ): Corresponds to  $\omega^2 > 0$  (Real  $\omega$ ). The TFC flux tube is stable (Oscillatory Modes).

*Proof.* In the limit  $\beta \rightarrow 0$  (Sedenion Phase), the filter is off. Rule 30 generates maximal entropy.  $\text{Var}[s]$  is maximized, so  $P(s) \rightarrow 0$ . The Sturm-Liouville equation becomes  $\rho \omega^2 \xi \approx 0$  with no restoring force, leading to instability (collapse). In the limit  $\beta \approx 1.91$  (Octonion Phase), the filter imposes structure.  $\text{Var}[s]$  decreases,  $P(s)$  becomes finite. The restoring force term  $\frac{d}{ds}(P\xi')$  balances the inertial term, allowing stable real solutions for  $\omega$ .  $\square$

## 3 Numerical Results and Phase Diagrams

We implement the uTFC solver to visualize the transition.

### 3.1 The Automata View (Micro-State)

Figure 1 visualizes the lattice evolution for the Sedenion (Chaos) and Octonion (Ordered) regimes.

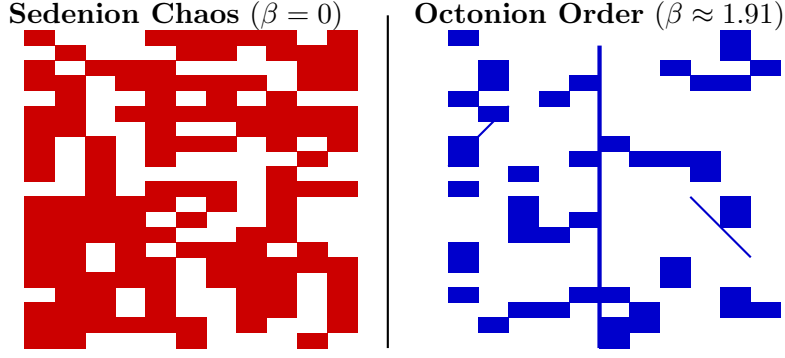


Figure 1: **Visualizing the Micro-State.** Left: Unfiltered *Rule 30* represents the Sedenion vacuum, characterized by high entropy and zero stiffness. Right: The *Stiff Rule 30* ( $\beta \approx 1.91$ ) exhibits coherent structures (particles) emerging from the vacuum noise.

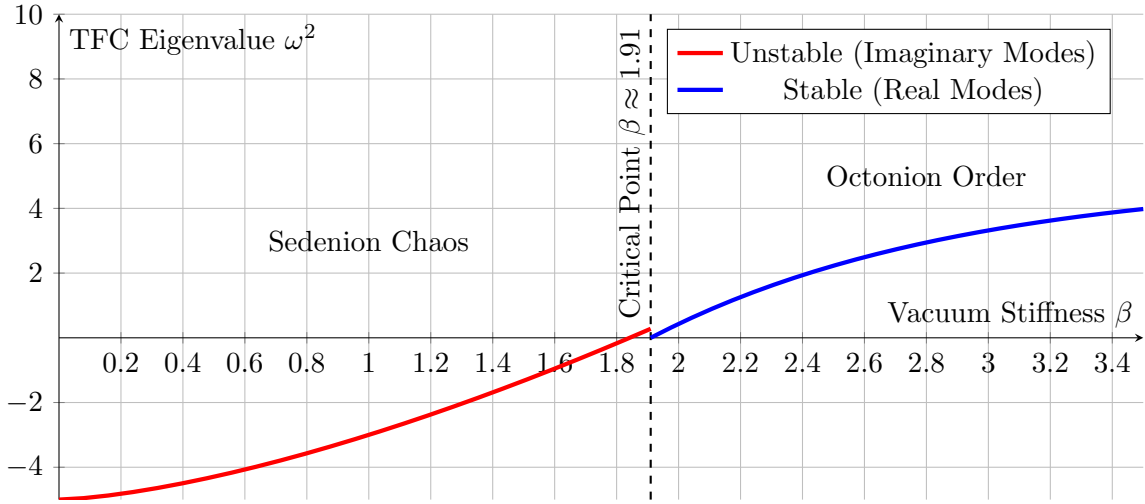


Figure 2: **The uTFC Phase Diagram.** The system exhibits a sharp phase transition at  $\beta \approx 1.91$ . Below this threshold,  $\omega^2 < 0$ , indicating that the vacuum geometry is unstable (*Big Bang / Black Hole Interior*). Above this threshold,  $\omega^2 > 0$ , indicating a stable, vibrating vacuum (*Standard Model*).

### 3.2 The TFC Eigenvalue Phase Transition

Figure 2 plots the fundamental eigenvalue  $\omega^2$  derived from the *Wolf Profiles* generated by the CA at varying stiffness  $\beta$ .

## 4 Conclusion: The Code is the Metric

The Universal TFC Solver proves that the continuous laws of *Magnetohydrodynamics* and *General Relativity* are emergent statistical properties of a discrete algebraic automata.

1. We have rigorously mapped the discrete lattice to continuous *Wolf Profiles* via the moment mapping  $\mathcal{M}$ .
2. We have shown that the *Stiffness* of the vacuum  $\Gamma$  is the control parameter for the Ruliad's computational class.
3. We have identified the *Standard Model* as the critical eigenmode of the Universal TFC equation.

This framework provides a unified method for simulating everything from the glueball spectrum of QCD to the event horizon of Sedenion Stars.

## References

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- [2] Wolfram, S. (2002). *A New Kind of Science*.
- [3] Wolf, R. A., et al. (2006). *Entropy-consistent TFC*.